A First Lean Formalization Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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Abstract

This note records the present state of a Lean formalization of a narrow part of Mochizuki's inter-universal Teichmuller theory: the passage from IUT III, Theorem 3.11 to Corollary 3.12, with attention to the Scholze–Stix objection. The current Lean code proves conditional route theorems, obstruction lemmas, and the final ordered-real algebra in Step (xi-f). It does not prove Corollary 3.12 and does not refute it. The main formal result so far is that a supplied hull/log-volume q-to-Θ comparison gives $C_\Theta \geq -1$, and that a nonarchimedean (Ind3) entry together with explicit Hodge/SHE square-weight transport data, or the primitive factored square/full-label SHE obligations, feeds such a comparison, while several simplified collapse mechanisms are rejected by Lean.

1 Scope

The review used the local Markdown conversions of IUT I–IV, Mochizuki's 2026 formalization note, and the Scholze–Stix note. The relevant source passages are:

- IUT I: initial Θ-data, prime strips, Hodge theaters, and non-scheme-theoretic Θ-links.
- IUT II: Hodge–Arakelov evaluation, theta values of shape q^{j^2}, multiradiality, and Gaussian monoids.
- IUT III: Theorem 3.11; Corollary 3.12, especially Steps (x) and (xi); Remarks 3.12.2 and 3.12.3.
- IUT IV: later explicit log-volume estimates and Diophantine consequences.
- Scholze–Stix: the ordered real-line identification problem in their Section 2.2.
- Mochizuki 2026: the "3.11.5 ⇒ 3.12" fourth-triangle decomposition.

2 Mathematical Target

IUT III Step (x) treats procession-normalized log-volumes as invariants for (Ind1) and (Ind2), and turns (Ind3) into an upper inequality. Step (xi) then uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot log-volume in an upper ray controlled by the Θ-pilot log-volume:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|} \quad \Rightarrow \quad -|\log(q)| \leq -|\log(\Theta)|.$$

If C_Θ is any real number satisfying

$$-|\log(\Theta)| \le C_\Theta |\log(q)|, \qquad |\log(q)| > 0,$$

then Step (xi-f) concludes $C_\Theta \ge -1$. This last inference is now a Lean theorem. Mochizuki's 2026 note isolates the final comparison as:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The present formalization targets only this corridor, not the full construction of Theorem 3.11.

3 Dispute Interface

Scholze–Stix require explicit bookkeeping for all ordered one-dimensional real vector spaces. Their pressure point is that the Θ-side concrete objects carry j^2-scaled degrees, while a consistent identification of the real lines appears to leave no room for inserting all these scalars.

The formalization therefore separates:

$$\begin{array}{c} \text{representative } j^2 \text{ data} \\ \text{balanced sign-compatible data} \\ \text{aggregate averaged data} \\ \text{final hull/log-volume data} \end{array} \quad \text{and} \quad \begin{array}{c} \text{raw real equality} \\ \text{transported equality} \\ \text{matched transport scales} \\ \text{cancellable ordered-line comparison.} \end{array}$$

This separation is the main correctness guard in the current code.

4 Lean Coverage

The relevant implementation is in

`Iut/Stage1/IUTStage1Source.lean` and `Iut/Stage1/IUTStage1Experiments.lean`.

The code currently covers:

- labeled real-line copies and positive-scale transports;
- a finite $\mathbb{F}_\ell$-label model with representative square weights $j \mapsto j^2$;
- a representative pilot-degree profile $\deg(\Theta_j) = j^2 \deg(q)$;
- proof that one label-independent scale cannot absorb all representative j^2 scales in this model;
- proof that, for nonzero $\deg(q)$, one label-independent scale cannot account for all representative Θ_j-degrees;
- proof that the raw representative j^2 theta-degree rule does not descend to the sign quotient $|F_\ell| = F_\ell/\{\pm 1\}$ when $\deg(q) \ne 0$;
- an absolute-label exponent on $|F_\ell| = \{0\} \cup F_\ell^\times/\{\pm 1\}$ which agrees with j^2 for the canonical representatives $0 \le j \le (\ell-1)/2$, and the corresponding theta-degree profile; Lean proves that the coordinate map $F_\ell \to |F_\ell|$ is surjective, that the zero absolute label is reached exactly by $j = 0$, hence has singleton coordinate fiber, and that every nonzero sign-label class has a nonzero coordinate representative; Lean also proves the exact fiber relation: two

coordinates have the same full absolute label if and only if they are equal or negatives of one another; equivalently, every nonzero full-label fiber is exactly a sign pair $\{j, -j\}$, hence has cardinality 2. Lean also proves that the half-range procession map to $|F_\ell|$ is bijective, using the minimal absolute representative $j_{\min}$ for surjectivity and the exponent j^2 for injectivity; this gives $\#|F_\ell| = (\ell+1)/2$ in the finite model and permits Lean to reindex absolute-label averages by half-range processions; Lean also proves that the half-range procession top is exactly $(\ell-1)/2$ and that the resulting finite index set has cardinality $(\ell+1)/2$, matching the $j \in \{0, \dots, (\ell-1)/2\}$ convention in IUT II; on this half-range procession, Lean proves the Gaussian degree formula $G(j) = j^2 \deg(q)$, with $G(0) = 0$, and the normalized average identity $6\ \mathrm{Avg}(G) = j_{\max}(2j_{\max}+1)\deg(q)$; by reindexing, the same identity holds for the average over $|F_\ell|$. Since $j_{\max} \geq 2$, Lean proves that this full average is not the singleton $j = 1$ identity value when $\deg(q) \neq 0$; in the nonpositive environment branch, the full $|F_\ell|$-average is bounded above by $\deg(q)$, and hence by any bound above $\deg(q)$; if $\deg(q) < 0$, this bound is strict: the full average is $< \deg(q)$, while for $\deg(q) > 0$ it is $> \deg(q)$; the same facts are now packaged as a typed $|F_\ell|$-indexed procession-normalized average object, where the average is provably distinct from the canonical $j = 1$ component unless $\deg(q) = 0$, strictly below it if $\deg(q) < 0$, strictly above it if $\deg(q) > 0$, and equal to that component if and only if $\deg(q) = 0$;

- the degree-level Gaussian evaluation $q \mapsto (q^{j^2})_{j \in |F_\ell|}$, including identity at $j = 1$; Lean now also converts such an evaluation into a cusp/full-label log-volume compatibility object and proves the full-label value-preservation branch when the source and target environment degrees agree; Lean also proves the IUT II restriction check at the canonical $j = 1$ sign label: the Gaussian component and the corresponding cusp-class log-volume are both the original environment degree; Lean also records the accompanying IUT II warning in the finite model: the singleton restriction to $j = 1$ is not stable under the additive F_ℓ-torsor action, since translation by 1 sends 1 to $2 \neq 1$; more generally, Lean proves that every nonempty translation-stable finite subset of F_ℓ is all of F_ℓ, so no nonempty proper subset restriction is compatible with this symmetry; Lean also proves that no nonzero additive translation descends to a well-defined map on $|F_\ell|$, since the equal absolute labels of t and $-t$ would have to map simultaneously to the nonzero label of $2t$ and to 0; equivalently, an additive translation descends to $|F_\ell|$ if and only if it is the zero translation; the same iff is exposed at the route interface as the exact condition for a translation equivalence to preserve full labels;

- a canonical representative square-weight profile on F_ℓ: Lean proves the total j^2-mass is positive from the $j = 1$ contribution and therefore constructs the profile without an external positivity premise; it also proves $6 \sum_{j \in \mathbb{Z}/\ell\mathbb{Z}} j_{\mathrm{rep}}^2 = (\ell-1)\ell(2(\ell-1)+1)$, hence the corresponding average formula after division by ℓ;

- the square-weighted constant-average check: a constant procession-normalized log-volume remains equal to the same constant after averaging with the canonical j^2-weights;

- the same constant-average check inside the $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$ cusp-label corridor: if the full-label log-volume is constant, the square-weighted average used in the Θ-route is that constant, including for the canonical weights;

- the constant-target structured-SHE subcase: the formal route now proves equality of the square-weighted average with the Θ-source average, not merely an upper bound;

- a log-volume shadow of the IUT III splitting-monoid action: the generator attached to a procession label contributes $j^2 \deg(q)$, contributes 0 at the core label, and adds its finite generator sum to the tensor-packet log-volume; Lean also proves the procession-normalized form: the acted tensor-packet average is the original tensor-packet average plus the averaged Gaussian generator contribution, equivalently the average of the $j^2 \deg(q)$ contributions over the procession container; for $S_{j+1} = \{0, \ldots, j\}$, Lean expands the total generator contribution to $j(j+1)(2j+1)\deg(q)/6$, and the resulting normalized log-volume change to $j(2j+1)\deg(q)/6$; since $j = \lfloor \ell/2 \rfloor \geq 2$, a nonnegative environment degree makes the acted normalized tensor-packet log-volume an upper bound for the original one;

- the finite procession containers $S^{\pm}_{j+1} = \{0, \ldots, j\}$, their nested inclusions, core-label stability, stage cardinality $j+1$, and total procession ambiguity $(\ell^{\pm})!$ rather than $(\ell^{\pm})^{\ell^{\pm}}$;

- the map from the final procession container to $|F_\ell|$, sending the core label to 0 and recovering the exponent j^2;

- the log-volume shadow of the tensor packet over $S^{\pm}_{j+1}$, with each factor a finite-fiber direct sum over places above $v_{\mathbb{Q}}$, total log-volume given by the finite sum over the container, normalized denominator $j+1$, and invariance under container permutations;

- the F_ℓ-label average specialization: for labels represented as $\mathbb{Z}/\ell\mathbb{Z}$, Lean rewrites the procession-normalized average denominator as ℓ;

- the zero/nonzero Gaussian average split: Lean proves

$$\#(\mathbb{Z}/\ell\mathbb{Z})^\times = \ell - 1, \qquad \mathrm{Avg}_{j \in F_\ell} G(j) = \frac{\ell - 1}{\ell} \mathrm{Avg}_{j \neq 0} G(j),$$

  for the degree-level Gaussian evaluation G, using $G(0) = 0$. Thus the paper's uniform $1/\ell$ weighted-volume convention is not silently replaced by the auxiliary nonzero-only average; in the negative signed branch Lean also proves the strict inequality $\mathrm{Avg}_{j \neq 0} G(j) < \mathrm{Avg}_{j \in F_\ell} G(j)$, while in the positive branch the inequality is reversed; equality between the two averages holds if and only if $\deg(q) = 0$. Lean also proves that the all-F_ℓ coordinate average itself is zero if and only if $\deg(q) = 0$; Lean separately formalizes the nonzero half-range $j = 1, \ldots, (\ell-1)/2$ average, proves that the signed nonzero carrier $F_\ell^\times$ is the twofold cover of this half-range by j and $-j$, and therefore proves that the signed nonzero average equals the nonzero absolute-label average. It also proves $6\,\mathrm{Avg}_{j \neq 0, |F_\ell|} G(j) = (j_{\max} + 1)(2j_{\max} + 1)\deg(q)$; Lean also proves the exact rescaling

$$\mathrm{Avg}_{|F_\ell|} G = \frac{j_{\max}}{j_{\max} + 1} \mathrm{Avg}_{j \neq 0, |F_\ell|} G.$$

  Since the full $|F_\ell|$-average has coefficient $j_{\max}(2j_{\max} + 1)/6$, Lean proves that the nonzero absolute-label average is strictly below the full average when $\deg(q) < 0$ and strictly above it when $\deg(q) > 0$, with equality if and only if $\deg(q) = 0$. Combining the signed and absolute computations, Lean proves

$$\mathrm{Avg}_{F_\ell} G = \frac{j_{\max}(j_{\max} + 1)}{3} \deg(q),$$

  equivalently $\mathrm{Avg}_{F_\ell} G = ((\ell+1)/\ell)\,\mathrm{Avg}_{|F_\ell|} G$, so $\mathrm{Avg}_{F_\ell} G$ is strictly below $\mathrm{Avg}_{|F_\ell|} G$ when $\deg(q) < 0$, strictly above it when $\deg(q) > 0$, and equal to it only when $\deg(q) = 0$;

- the IUT II Remark 4.7.3 zero/nonzero weighted-volume relation: Lean records a relation in which every nonzero full label is subordinate to the zero label, proves that $0 \ll 0$ is false, proves that this relation is not symmetric at the zero label, and proves the exact criterion $a \ll 0 \iff a \neq 0$ for full labels; at coordinate level this becomes $j \ll 0 \iff j \neq 0$, so the all-coordinate statement is false precisely because it includes $j = 0$. Lean also proves that the full-label set subordinate to zero has cardinality $(\ell - 1)/2$, the nonzero absolute half-range, and decomposes any full-label sum, hence any full-label average, into its zero contribution plus the subordinate nonzero contribution with denominator $(\ell + 1)/2$. Applying this to the Gaussian degree function, whose zero component is 0, Lean rewrites the full absolute-label Gaussian average as the subordinate nonzero Gaussian sum divided by the full absolute-label denominator and proves the closed form

  $$6 \sum_{a \ll 0} G(a) = j_{\max}(j_{\max} + 1)(2j_{\max} + 1) \deg(q).$$

  Lean also proves that this subordinate sum is exactly the canonical nonzero absolute half-range sum $\sum_{j=1}^{j_{\max}} G(j)$. In contrast to nonzero additive translations, Lean defines the multiplicative unit action on full labels, proves its compatibility with the coordinate formula $j \mapsto a \cdot j$, and proves that it preserves the subordinate zero/nonzero relation. Lean also proves the unit and multiplication laws for this action and proves that the sign subgroup $\{\pm 1\}$ acts trivially on full absolute labels. Each unit action is packaged as an equivalence of full labels with inverse action by a^{-1}, and Lean proves that the zero label is fixed exactly. Consequently Lean proves that full-label sums and full-label averages are invariant under this multiplicative permutation; the same holds for sums over the subordinate nonzero full-label subset. These invariance statements are also specialized to the Gaussian degree function $G(a)$, so the full Gaussian average and the subordinate Gaussian sum are invariant under multiplicative reindexing. At the coordinate level, Lean proves the corresponding sharp contrast: unit multiplication preserves the predicate $j \ll 0$, whereas additive translation preserves this predicate for all j if and only if the translation is zero. More strongly, an affine coordinate operation $j \mapsto aj + t$, with $a \in F_\ell^\times$, descends to full absolute labels if and only if $t = 0$, and preserves the subordinate predicate for all j if and only if $t = 0$. Lean separately proves that raw F_ℓ-coordinate sums are invariant under additive translation, unit multiplication, and their affine composition; this is only finite-permutation invariance and does not supply descent to $|F_\ell|$. For the Gaussian coordinate average, Lean packages this as a diagnostic: every affine map with nonzero additive offset preserves the raw coordinate average but fails full-label descent. Lean also converts the zero/nonzero cusp-label compatibility object to a uniform F_ℓ average and proves that a diagonal constant log-volume distribution averages to that same constant;

- the IUT III Step (x) averaged-log-volume corridor: Lean records that (Ind1) and (Ind2) preserve the procession-normalized averaged log-volume, while (Ind3) supplies an upper bound; the before-, after-(Ind1)-, and after-(Ind2)-averages are all bounded by the same (Ind3) upper bound;

- the Step (x) to Step (xi) upper-ray handoff: if the q-pilot reading is the pre-indeterminacy averaged log-volume and the Θ-hull reading is the (Ind3) upper bound, Lean constructs the hull/determinant upper-ray datum used in Step (xi);

- the Step (x) to Step (xi-g) two-computation handoff: the same data constructs the q-pilot two-computation record, with the input prime-strip reading equal to the pre-indeterminacy average and bounded by the Θ-hull upper-ray value;

- the Step (x)-sourced C_Θ endpoint: adding $|\log(q)| > 0$ for the pre-indeterminacy q-pilot average and the displayed upper bound $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ yields $C_\Theta \geq -1$;

- the Step (x)-sourced statement endpoint: after adjoining the statement-level Θ-subject/q-not-subject boundary, the same data constructs the Corollary 3.12 statement endpoint, with finite Θ-branch excluding $+\infty$ and $C_\Theta \geq -1$; Lean proves that this constructed endpoint still records the Θ-pilot as subject to (Ind1), (Ind2), (Ind3) and the q-pilot as not subject to them, and that these two status fields are unequal at the constructed endpoint; Lean also proves that the endpoint q-pilot log-volume is exactly the pre-indeterminacy average and, by Step (x) invariance, also the after-(Ind1)- and after-(Ind2)-averages; thus $|\log(q)|$ is the negative of these equal averages and is positive; equivalently, the three equal averages are negative under the q-pilot positivity hypothesis; Lean also proves the displayed inequality $-|\log(q)| \leq -|\log(\Theta)|$ at this endpoint, with q read as the pre-indeterminacy average and Θ read as the finite real branch; equivalently, the endpoint q-pilot log-volume lies in the constructed upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$; Lean also formalizes the opening reduction in the printed proof: if this finite real Θ-value is nonnegative, then the strict negativity of the q-pilot log-volume gives the comparison immediately; moreover the displayed bound $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ then forces $C_\Theta \geq 0$; the finite extended Θ-value and the real Θ-branch are identified with the Step (x) (Ind3) upper bound; the same endpoint bound holds after the (Ind1) and (Ind2) average-preserving transformations; the Step (x)-sourced endpoint also directly rejects $C_\Theta < -1$;

- base-valuation products of tensor packets, the finite $(F^\times_{\text{mod}})_j$-action shadow, and a coric transfer shadow $F \to D \to D^\vdash$ preserving product log-volume without identifying Hodge-theater histories;

- the Step (xi-d)–(xi-g) shadow: determinant normalization, upper-ray membership, and equality of the two q-pilot log-volume computations;

- the Corollary 3.12 finiteness endpoint: the statement-level value $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$ is represented by an explicit finite/$+\infty$ split, and the hull/determinant upper-ray datum forces the finite real branch, explicitly excluding the $+\infty$ branch; Lean also exposes the resulting real Θ-log-volume and proves the q-pilot upper-ray comparison against that real value;

- the Corollary 3.12 statement boundary: the Θ-pilot log-volume is typed as subject to (Ind1), (Ind2), (Ind3), while the q-pilot log-volume is typed as not subject to these indeterminacies;

- the extended statement endpoint: from the finite Θ-branch, $0 < -|\log(q)|$, the upper-ray comparison, and the displayed hypothesis $-|\log(\Theta)| \leq C_\Theta |\log(q)|$, Lean proves $C_\Theta \geq -1$;

- the Step (xi-f) final algebra: from $-|\log(q)| \leq -|\log(\Theta)|$, $|\log(q)| > 0$, and $-|\log(\Theta)| \leq C_\Theta |\log(q)|$, Lean proves $C_\Theta \geq -1$;

- the signed-payload bridge: the existing Stage 1 comparison record $q_{\text{signed}} \leq \Theta_{\text{signed}}$, together with $\Theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}})$, feeds this Step (xi-f) algebra directly;

- the Step (xi-g) endpoint bridge: the two-computation q-pilot record produces the signed comparison payload with q_{signed} read from the input prime-strip log-volume and Θ_{signed} read from the hull/determinant upper-ray boundary;

- the Step (xi-e)–(xi-f) fixed-value condition: in the signed two-computation endpoint, Lean proves that the input prime-strip reading and the hull-output reading are both the same real value $-|\log(q)|$, and that this value lies in the upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$;

- the fixed-value C_Θ endpoint: combining the preceding fixed upper-ray membership with $-|\log(\Theta)| \leq C_\Theta|\log(q)|$, Lean derives $-|\log(q)| \leq C_\Theta|\log(q)|$ and hence $C_\Theta \geq -1$; consequently the same hypotheses reject any $C_\Theta < -1$;

- the statement-endpoint composition: the two-computation C_Θ endpoint canonically supplies the finite Θ-branch Corollary 3.12 statement endpoint once the Θ-subject/q-not-subject indeterminacy boundary is provided;

- the opening sign reduction in the proof of Corollary 3.12: from $0 < -q_{\text{signed}}$, Lean proves that $0 \leq \Theta_{\text{signed}}$ immediately implies $q_{\text{signed}} \leq \Theta_{\text{signed}}$, matching the printed reduction to the case $-|\log(\Theta)| < 0$; the same strict-negativity argument is now available at the statement endpoint for the real Θ-log-volume;

- the Step (xi-h) warning: for $N \geq 2$ and $\Theta < 0$, the tensor-power boundary satisfies $N\Theta < \Theta$;

- the Step (xii) shadow: local Frobenioid p_v^N-ambiguity changes log-volume by an integer shift, while global calibration fixes exponent 0;

- the Remark 3.12.1(iii) arithmetic fact $\mathbb{Q}_{>0} \cap \mathbb{Z}^\times = \{1\}$;

- the Remark 3.12.1(ii) generalized-link numerical bound $C_\Theta \geq -\lambda$ for $\lambda \in \mathbb{Q}_{>0}$, with strict strengthening over the $\lambda = 1$ bound when $\lambda < 1$;

- the q^λ-pilot endpoint algebra: from a signed comparison for $-\lambda|\log(q)|$ and the same $C_\Theta|\log(q)|$ upper bound, Lean derives $C_\Theta \geq -\lambda$, and hence $C_\Theta \geq -1$ when $\lambda \leq 1$;

- the Remark 3.12.2 distinct-label implication q-itw $\Rightarrow$ q-itw $\wedge$ Θ-itw/indets, with q-native and Θ-native labels proved distinct;

- the Remark 3.12.2 toy calculation: forgotten concrete assignments $-h$ and $-2h$ are incompatible for $h > 0$, while $-h \in \mathbb{R}_{\leq -2h+\epsilon}$ implies $h \leq \epsilon$;

- the Fig. 3.9 column table: both $\bullet_0$ and $\circ$ roles are required in both columns, while q-pilot multiradiality is absent;

- the Remark 3.12.2(v) absorption distinction: zero-column unit indeterminacy is absorbed by a hull upper ray, while one-column logarithmic conjugacy is absorbed as log-volume equality;

- the Remark 3.12.3(i) metric analogy: $\mathrm{vol}(S) > 0$ and $\chi_S = -\mathrm{vol}(S)$ imply $\chi_S < 0$;

- the Remark 3.12.4(iii) factor-p analogy: theta-label normalization divides a gap log-volume by ℓ, and multiplication by ℓ recovers it;

- the Remark 3.12.4(v) p-adic analogue: an inclusion, not isomorphism, gives $(1-p)(2g-2) \leq 0$;

- the IUT IV, Theorem 1.10 algebraic handoff: an explicit expression for C_Θ, together with $C_\Theta \geq -1$ and the remaining elementary estimates, implies the displayed $\frac{1}{6}\log(q)$ bound;

- the final elementary coefficient estimates in IUT IV, Theorem 1.10: $(1 - 12/\ell^2)^{-1} \leq 2$ and $(1 - 12/\ell^2)^{-1}\frac{1}{4}(1 + 36d_{\text{mod}}/\ell) \leq 1 + 80d_{\text{mod}}/\ell$;

- the IUT IV, Theorem 1.10 base-change monotonicity: $\log(d_{F_{\mathrm{tpd}}}) + \log(f_{F_{\mathrm{tpd}}}) \leq \log(d_F) + \log(f_F)$ propagates through the final right-hand side;
- the IUT IV, Theorem 1.10 Step (i) identities: $n^{-1}\sum_{m=1}^{n} m = (n+1)/2$ and $n^{-1}\sum_{m=1}^{n} m^2 = (2n+1)(n+1)/6$;
- the IUT IV, Theorem 1.10 Step (ii) small-prime error: $\log(2^{11}3^3 5^2) \leq 21$, represented by the corresponding linear logarithmic bound;
- the IUT IV, Theorem 1.10 Step (i) GL_2 arithmetic constants for $p = 2, 3, 5$, including the product with the quadratic factor from (E5);
- the IUT IV, Theorem 1.10 Step (iii) algebraic estimate combining the Step (ii) K-bound and the displayed $\log(s_Q)$-bound into $(1 + 4/\ell)\log(d_K, f_K) + (16/\ell)\log(s_Q) \leq (1 + 36d_{\mathrm{mod}}/\ell)S + 2\log(\ell) + 70$;
- the IUT IV, Theorem 1.10 Step (viii) Proposition 1.6 case split: the two cases $e^*_{\mathrm{mod}}\ell \geq \eta_{\mathrm{prm}}$ and $e^*_{\mathrm{mod}}\ell < \eta_{\mathrm{prm}}$ imply the uniform $4/3(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$ bound;
- the IUT IV, Theorem 1.10 Step (viii) final absorption of $2\log(\ell) + 74$ and the prime-product term into $10(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$;
- the IUT IV, Theorem 1.10 final display: the Corollary 3.12 C_Θ handoff, followed by the tripodal-to-F monotonicity, gives the final F-level $\frac{1}{6}\log(q)$ inequality;
- the IUT IV, Theorem A bounded-discrepancy conclusion: boundedness below of $(1 + \epsilon)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) - \mathrm{ht}_{\omega_X(D)}$ implies the corresponding pointwise height inequality on points of bounded degree;
- the IUT IV, Corollary 2.2(i) bounded-discrepancy chain: $\frac{1}{6}\log(q_2) \approx \frac{1}{6}\log(q_\forall) \approx \frac{1}{6}\mathrm{ht}_\infty \approx \mathrm{ht}_{\omega_X(D)}$ composes to a direct bounded discrepancy relation;
- bounded-discrepancy transfer lemmas: pointwise upper and lower bounds move across an $\approx$ relation with the expected additive constant shift;
- the IUT IV, Corollary 2.2(ii) condition (C1) prime-scale window: $(\log(q_\forall))^{1/2} \leq \ell \leq 10\delta(\log(q_\forall))^{1/2}\log(2\delta\log(q_\forall))$. Lean keeps the square-root and logarithmic factor explicit, proves $\log(q_\forall) \geq 0$, and derives $1 \leq 10\delta\log(2\delta\log(q_\forall))$ from the nonempty window;
- the IUT IV, Corollary 2.2(ii) use of (C1) in Theorem 1.10: $\sqrt{h} \leq \ell$ and $80d_{\mathrm{mod}} \leq \delta$ imply $80d_{\mathrm{mod}}/\ell \leq \delta h^{-1/2}$;
- the IUT IV, Corollary 2.2(ii) error-term use of (C1): $d^*_{\mathrm{mod}} \leq \delta$ and $\ell \leq 10\delta h^{1/2}\log(2\delta h)$ imply $20(d^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}}) \leq 200\delta^2 h^{1/2}\log(2\delta h) + 20\eta_{\mathrm{prm}}$;
- the IUT IV, Theorem 1.10 to Corollary 2.2(ii) first-bound composition: these two C1 replacements convert the Theorem 1.10 right-hand side into $(1 + \delta h^{-1/2})S + 200\delta^2 h^{1/2}\log(2\delta h) + 20\eta_{\mathrm{prm}}$;
- the IUT IV, Corollary 2.2(ii) q_2/q comparison: the displayed estimate $\frac{1}{6}\log(q_2) - \frac{1}{6}\log(q) \leq \frac{1}{3}h^{1/2}\log(2\delta h)$ follows through the intermediate term $h^{1/2}\log(\ell)$;

- the IUT IV, Corollary 2.2(ii) pre-ϵ_E bound for $h = \log(q_\forall)$: the Theorem 1.10 bound, the q_2/q estimate, and the bounded discrepancy B_K imply

$$\frac{1}{6}h \leq (1 + \delta h^{-1/2})S + (15\delta)^2 h^{1/2} \log(2\delta h) + \frac{1}{2}C_K,$$

  with $C_K = 40\eta_{\mathrm{prm}} + 2B_K$;

- the IUT IV, Corollary 2.2(ii) condition (C2) inequality chain: $\frac{1}{6}\log(q) \leq \frac{1}{6}\log(q_2) \leq \frac{1}{6}\log(q_\forall) \leq (1+\epsilon_E)(\log\operatorname{diff}_X + \log\operatorname{cond}_D) + C_K$;

- the IUT IV, Corollary 2.2(ii) definition of ϵ_E: $\epsilon_E = (60\delta)^2 h^{-1/2}\log(2\delta h)$, represented with $h^{1/2}$ and $\log(2\delta h)$ as explicit real parameters; Lean proves positivity and $\epsilon_E \geq 5\delta h^{-1/2}$ from $\delta \geq 2$, $h^{1/2} > 0$, and $\log(2\delta h) \geq 1$;

- the IUT IV, Corollary 2.2(ii) ϵ_E-error conversion: substituting the definition of ϵ_E, Lean proves $(15\delta)^2 h^{1/2}\log(2\delta h) \leq \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$;

- the IUT IV, Corollary 2.2(ii) bridge to the denominator step: the pre-ϵ_E h-bound, the error conversion, and $\delta h^{-1/2} \leq \epsilon_E/5$ yield the preliminary inequality used to move the $\frac{2}{5}\epsilon_E\frac{1}{6}h$ term to the left;

- the IUT IV, Corollary 2.2(ii) denominator step: Lean moves the term $(2\epsilon_E/5)\frac{1}{6}h$ to the left and obtains the denominator $(1 - 2\epsilon_E/5)^{-1}$;

- the IUT IV, Corollary 2.2(ii) final ϵ_E-absorption: from $0 < \epsilon_E \leq 1$, Lean proves

$$\frac{1 + \epsilon_E/5}{1 - 2\epsilon_E/5} \leq 1 + \epsilon_E, \qquad (1 - 2\epsilon_E/5)^{-1} C_K/2 \leq C_K,$$

  and hence the displayed C2 bound from the preceding denominator expression;

- the IUT IV, Corollary 2.2(ii) final h-bound: Lean composes the pre-ϵ_E bound, error conversion, move-left step, and absorption to prove $\frac{1}{6}h \leq (1+\epsilon_E)S + C_K$;

- the IUT IV, Corollary 2.2(ii) final C2 bridge: using $S \leq \log\operatorname{diff}_X(x_E) + \log\operatorname{cond}_D(x_E)$, Lean derives the displayed C2 inequality chain;

- the IUT IV, Corollary 2.2(ii) comparison $\log\operatorname{diff}_X(x_E) = \log(d_{F_{\mathrm{tpd}}})$ and $\log(f_{F_{\mathrm{tpd}}}) \leq \log\operatorname{cond}_D(x_E) \leq \log(f_{F_{\mathrm{tpd}}}) + \log(2\ell)$;

- the IUT IV, Corollary 2.2 to Theorem A passage: bounded-discrepancy equivalence $\frac{1}{6}\log(q_2) \approx \mathrm{ht}_{\omega_X(D)}$, together with the C2 bound for $\frac{1}{6}\log(q_2)$, gives a lower bound for $(1+\epsilon_E)(\log\operatorname{diff}_X + \log\operatorname{cond}_D) - \mathrm{ht}_{\omega_X(D)}$;

- the IUT IV, Corollary 2.2/2.3 finite-exception passage: a lower bound on the complement of a finite exceptional set and a lower bound on the exceptional set glue to a global lower bound by taking the minimum of the two constants;

- the IUT IV, Corollary 2.3 Diophantine inequality: Lean records the GenEll transfer as an explicit hypothesis and derives the global bounded-discrepancy lower bound for $(1 + \epsilon)(\log\operatorname{diff}_X + \log\operatorname{cond}_D) - \mathrm{ht}_{\omega_X(D)}$;

- comparison levels: pointwise representative, aggregate representative, balanced sign-compatible, and hull/log-volume;
- endpoint audits for the $3.11.5 \Rightarrow 3.12$ comparison chart;
- nonarchimedean and archimedean (Ind3) local orientations;
- ordered real-line alignment with cancellation only after scale equality;
- a canonical unit-scale ordered alignment from a nonarchimedean (Ind3) entry;
- the local-object Hodge-descent square-weight boundary: transported zero-column and cusp-class local objects have the same finite log-volume as the packet object, while square/full-label preservation obligations remain explicit;
- the square/full-label Hodge/SHE transport audit: a pointwise upper bound on the transported target full-label log-volumes transfers to the source square-weighted average once full-label log-volume, square weight, and total-weight preservation are supplied; the same transfer is now available inside the structured-SHE wrapper, and the resulting upper bound is exposed for the square-weighted average in the F_ℓ cusp-label route after matching the source profile and source log-volume; consequently, structured-SHE target full-label bounds by the Θ-source average feed the cusp-class container route all the way to $q_{\text{signed}} \leq \Theta_{\text{signed}}$; the same final comparison is now available directly from the primitive factored square/full-label SHE obligations, without requiring a prebuilt transport-audit argument;
- a Gaussian-to-factored-SHE bridge: source and target degree-level Gaussian evaluations with equal environment degree supply the full-label value-preservation branch of the factored square/full-label SHE obligations; with separate coordinate-square and full-label-map preservation, the experiment route reaches $q_{\text{signed}} \leq \Theta_{\text{signed}}$ directly from Gaussian target bounds; since the absolute-label exponent is nonnegative, Lean also derives those target bounds in the branch where the target environment degree is nonpositive and the Θ-source average is nonnegative; conversely, the all-label Gaussian target-bound formulation forces the Θ-source average to be nonnegative, since the core label has Gaussian degree 0; after removing the core label, Lean proves that every nonzero Gaussian target value is bounded by the Θ-source average if the target environment degree is nonpositive and is itself bounded by that average; Lean also forms the corresponding finite average over the nonzero $ZMod$ carrier, proves the same upper bound for this nonzero average, and proves the exact rescaling between this auxiliary nonzero average and the uniform F_ℓ-average with zero Gaussian term; because the square-weighted route assigns weight 0 to the core label, Lean now feeds nonzero-only target bounds through the square/full-label transport audit all the way to $q_{\text{signed}} \leq \Theta_{\text{signed}}$; this nonzero Gaussian environment-comparison route is now a source-layer theorem, not only an experiment wrapper;
- a bridge from factored square/full-label SHE obligations to the Hodge-descent (Ind3) route.

5 Current Lean Results

The current first-pass experiment report evaluates to:

$$
\begin{array}{ll}
\text{route theorem flags} & = \texttt{true}, \\
\text{collapse-obstruction flags} & = \texttt{true}, \\
\text{Gaussian/procession finite-model flags} & = \texttt{true}, \\
\texttt{balancedLevelRejectedAtFinalRouteTheoremAvailable} & = \texttt{true}, \\
\texttt{disputeSettledByCurrentStage} & = \texttt{false}.
\end{array}
$$

Here the route flags are ordered real-line, nonarchimedean entry, and factored SHE availability. The obstruction flags are raw-cancellation failure, label-independent j^2 failure, and sign-quotient representative descent failure. The Gaussian/procession flags include finite tensor-packet, monoid action, normalized splitting-generator average, the explicit square-sum formula for the generator total and normalized delta, the resulting conditional monotonicity of the acted tensor packet, canonical positive square-weight profile, the constant-average check, base-valuation product, coric-transfer, determinant-normalization, upper-ray, q-pilot two-computation, the Step (x) averaged-log-volume indeterminacy corridor, the refined Step (x) bounds for the before-, after-(Ind1)-, and after-(Ind2)-averages, tensor-power warning, and global Frobenioid-calibration shadows, plus the finiteness endpoint for $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$, including extraction of the finite real Θ-value, the direct Corollary 3.12 statement endpoint with Θ-subject/q-not-subject pilot bookkeeping, the Step (x)-sourced upper-ray, q-pilot two-computation, statement endpoint, propagation of the same pilot-status boundary to that endpoint, proof that the endpoint Θ- and q-statuses are distinct, identification of its q-pilot real with the Step (x) pre-, after-(Ind1)-, and after-(Ind2)-averages, positivity of $|\log(q)|$ at that endpoint, and direct $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$, $-|\log(q)| \leq -|\log(\Theta)|$, finite-$\Theta$ exclusion of $+\infty$, identification of the finite extended Θ-value with the Step (x) (Ind3) upper bound, the nonnegative-Θ opening reduction including the stronger $C_\Theta \geq 0$ branch, and $C_\Theta \not< -1$ conclusions, the Step (xi-g)-to-(xi-f) endpoint bridge giving $C_\Theta \geq -1$ from the two-computation q-pilot record, the Step (xi-e)–(xi-f) fixed $-|\log(q)|$ membership check and its direct $C_\Theta \geq -1$ consequence, the composition into the statement-level endpoint, and positive-rational unit rigidity. The generalized-link flag records the rational parameter inequality $C_\Theta \geq -\lambda$, now with the corresponding q^λ-pilot real-valued derivation. The distinct-label flag records the logical intertwining display in Remark 3.12.2 and rejects collapse of the q-native and Θ-native labels. The toy-model flag records the resulting upper-ray arithmetic $h \leq \epsilon$. The column-table flag records the Fig. 3.9 theta/q similarity-difference split. The absorption flag records the zero-column upper-bound versus one-column equality distinction. The metric flag records the Gauss–Bonnet sign calculation used as analogy for the upper semi-compatibility inequality. The factor-p flag records the F_ℓ-label averaging denominator. The Frobenius-derivative flag records the degree inequality from inclusion. The IUT IV flag records the downstream Theorem 1.10 use of the Corollary 3.12 lower bound on C_Θ, including the final coefficient estimates and the tripodal-to-F log-degree monotonicity. The Step (i) flag records the two elementary sum identities used in the leading-term computation. The Step (ii) small-prime flag records the additive error bound 21. The GL_2 flag records the finite cardinality constants feeding that bound. The Step (iii) flag records the $\log(s_Q)$ coefficient absorption. The Step (viii) flag records the prime-product case split used before the final upper bound for C_Θ, plus the final coarse error absorption. The final display flag records the composed Theorem 1.10 inequality with F-level log-degree terms. The Theorem A flag records the bounded-discrepancy height/log-different/log-conductor inequality. The Corollary 2.2 flag records composition of the displayed bounded-discrepancy chain. The transfer flag records movement of pointwise bounds across bounded-discrepancy equivalence. The Corollary 2.2(C1) flag records the displayed prime-scale window and its elementary consequences. The Corollary 2.2(C2) flag records the displayed three-step logarithmic inequality chain. The C1 coefficient flag records the replacement of $80d_{\mathrm{mod}}/\ell$ by $\delta h^{-1/2}$ in the Theorem 1.10-to-C2 proof. The C1 error-term flag records the corresponding replacement of $20d^*_{\mathrm{mod}}\ell$ by $200\delta^2 h^{1/2}\log(2\delta h)$. The first-bound flag records the composition of those replacements with Theorem 1.10. The ϵ_E-definition flag records the source formula and lower estimate for ϵ_E. The ϵ_E-move-left flag records introduction of the denominator $(1-2\epsilon_E/5)^{-1}$. The ϵ_E-absorption flag records the final denominator removal in C2. The ϵ_E-error flag records the conversion of the $(15\delta)^2$-term to the $\frac{2}{5}\epsilon_E\frac{1}{6}h$ term. The final h-bound flag records the full composition through absorption. The final C2 bridge flag records the last monotonic replacement of the tripodal log-degree sum by the curve different/conductor sum. The h-to-move-left flag records the bridge from the pre-ϵ_E bound to the denominator-stage preliminary inequality. q2/q flag records the displayed comparison of $\log(q_2)$ and $\log(q)$. The pre-ϵ_E h-bound flag records the subsequent absorption of the bounded $q_\forall/q_2$ discrepancy into C_K. The log-different/log-conductor flag records the bridge from tripodal field terms to curve functions. The Corollary 2.2-to-Theorem A flag records the formal bounded-discrepancy transfer from C2 to the Theorem A inequality. The finite-exception flags record the lower-bound gluing used when finite exceptional sets are enlarged in Corollary 2.2 and then passed toward Corollary 2.3. The Corollary 2.3 flag records the final GenEll-transfer interface for the Diophantine inequality. The theorem-level route is:

$$\begin{array}{ccccc}
\text{nonarch. (Ind3) entry} & \to & \text{canonical ordered } \mathbb{R}\text{-alignment} & \to & \text{source/target alignment} \\
 & & \downarrow & & \downarrow \\
\text{factored SHE transport} & \to & \text{Hodge-descent audit} & \to & q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}}.
\end{array}$$

The route remains conditional on the displayed hypotheses. The local-object Hodge-descent layer now exposes equality of the transported finite log-volumes before square/full-label preservation is supplied. The square/full-label audit now also transports target pointwise upper bounds back to the source square-weighted average, including through the structured-SHE route wrapper, and the experiment layer exposes the resulting final q-to-Θ comparison theorem. The same experiment-level theorem is now exposed for primitive factored square/full-label SHE obligations and for degree-level Gaussian evaluations after the explicit preservation branches are supplied. In the nonpositive-environment/nonnegative-Θ-average branch, the Gaussian pointwise target bound is derived from the sign of the q^{j^2}-exponent rather than assumed. Lean also proves that this all-label Gaussian bound cannot hold when the Θ-source average is negative. For nonzero labels, Lean proves the complementary bound: the absolute exponent is at least 1, so a nonpositive environment degree below the Θ-source average bounds every nonzero Gaussian component and hence their finite nonzero-label average. The square-weighted transport route has also been strengthened from an all-label pointwise bound to a weighted-summand bound, so the zero label requires no target upper bound. The experiment theorem now delegates to the source-layer q-to-Θ theorem for this nonzero Gaussian environment-comparison route. Separately, Lean proves that the uniform F_ℓ-average of the Gaussian shadow is $(\ell-1)/\ell$ times the auxiliary nonzero-carrier average, fixing the denominator at ℓ when the zero label is present.

6 Audit Findings

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The strongest positive result is conditional: if the hull/log-volume upper-ray comparison supplies $-|\log(q)| \le -|\log(\Theta)|$, then Lean checks the formal Step (xi-f) conclusion $C_\Theta \ge -1$. If the nonarchimedean (Ind3) entry, ordered real-line alignment, and Hodge/SHE square-weight transport audit are supplied, Lean checks the final signed q-to-Θ inequality feeding that algebra. If the corresponding primitive factored square/full-label SHE obligations are supplied instead, Lean first constructs the transport audit and checks the same final inequality. If the input is a pair of degree-level Gaussian evaluations with equal environment degree, Lean constructs the full-label log-volume compatibility branch before invoking the factored route. The branch with nonnegative Θ-source average is formally easier: nonnegativity of j^2 makes every nonpositive environment-degree Gaussian target value bounded above by that average. The all-label version cannot be the negative branch: the core label contributes 0, so a negative upper bound is impossible. On the nonzero-label part, the formal obstruction disappears: exponent ≥ 1 gives the expected bound from the environment comparison. The square-weighted final route now accepts this nonzero-only bound, since the core summand has weight 0. This gives a conditional source-level q-to-Θ route from degree-level Gaussian data, coordinate-square preservation, full-label-map preservation, equal source/target environment degree, and environment degree bounded above by the Θ-source average. The zero/nonzero average split is now exact: the nonzero carrier has cardinality $\ell - 1 = 2j_{\max}$, the signed carrier is a twofold cover of the nonzero absolute half-range, the zero Gaussian degree is 0, and the all-F_ℓ average is $(\ell-1)/\ell$ times the nonzero absolute average; consequently the F_ℓ-coordinate and full $|F_\ell|$-absolute averages differ strictly for nonzero environment degree. The remaining task is to replace the current degree-level Gaussian shadow by the actual Hodge–Arakelov Gaussian/Frobenioid construction behind this environment comparison.

The strongest negative result is also limited: in the current $\mathbb{F}_\ell$-label model, Lean rejects a single label-independent scale accounting for all representative j^2 factors, and it rejects balanced sign-compatible evidence as the final hull/log-volume comparison level. Lean also rejects direct descent of arbitrary raw representative j^2 theta-degree data to $|F_\ell|$, while accepting the half-range exponent convention stated in IUT II. Lean also checks that an N-th tensor-power boundary is a strict strengthening for $N \geq 2$ and negative pilot log-volume. It also checks the Step (xii) numerical warning: nonzero local Frobenioid shifts change log-volume, whereas global calibration restores the unshifted value. The positive rational unit rigidity fact in Remark 3.12.1(iii) is formalized directly over $\mathbb{Q}$. For generalized links, Lean records that $\lambda < 1$ makes $-\lambda > -1$. Lean also checks that the Remark 3.12.2 simultaneous-intertwining step depends on distinct labels rather than an unlabeled identification. In the accompanying toy model, Lean proves the final upper bound exactly from the upper-ray membership hypothesis. Lean also records that the q-pilot column lacks the theta-side multiradiality used in Theorem 3.11(iii)(c). For Remark 3.12.2(v), Lean separates hull absorption of unit shifts from log-volume equality after logarithmic-conjugate compatibility. For Remark 3.12.3(i), Lean proves the stated negative Euler-characteristic sign from positivity of metric volume. For Remark 3.12.4(iii), Lean checks the ℓ-denominator normalization identity for the theta-label averaging analogy. For Remark 3.12.4(v), Lean proves the displayed nonpositive degree defect from $p \geq 2$ and $g \geq 2$. For IUT IV, Theorem 1.10, Lean proves the algebraic implication from the computed C_Θ expression and $C_\Theta \geq -1$ to the final $\frac{1}{6}\log(q)$ estimate, conditional on the separate elementary estimates. Lean now also proves the two displayed final elementary coefficient estimates from $\ell \geq 5$. It also proves the final monotonic replacement of the tripodal log-degree sum by the larger F-level log-degree sum, conditional on the two component monotonicity inequalities cited from Proposition 1.3(i). Lean also proves the Step (i) sum and square-sum identities over $\mathbb{R}$. For Step (ii), Lean proves the small-prime additive error bound from the displayed logarithmic inequalities. Lean also checks the Step (i) GL_2 constants for $p = 2, 3, 5$ and their product with the quadratic extension factor. For Step (iii), Lean proves the displayed coefficient absorption from the source hypotheses $d_{\mathrm{mod}} \geq 1$, $\ell \geq 5$, $2\log(\ell) \leq \ell$, and nonnegativity of the tripodal log-degree sum. For Step (viii), Lean checks that the two Proposition 1.6 cases imply the uniform prime-product bound used in the final C_Θ estimate. Lean also checks the subsequent absorption into the coefficient 10 of $e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}}$. Combining the Corollary 3.12 handoff with the tripodal-to-

F monotonicity, Lean proves the final displayed F-level inequality of Theorem 1.10 at this real-valued shadow level. For Theorem A, Lean records the exact bounded-discrepancy conclusion and proves the equivalent pointwise height bound from the lower-bound hypothesis. For Corollary 2.2(i), Lean defines equality up to bounded discrepancy, proves reflexivity, symmetry, transitivity, nonnegative scaling, and composes the displayed chain from $\log(q_2)$ to $\mathrm{ht}_{\omega_X(D)}$. Lean also proves that upper and lower pointwise bounds transfer across such a bounded-discrepancy equivalence with the appropriate constant shift. For Corollary 2.2(ii)(C1), Lean records the displayed window for ℓ and derives nonnegativity of $\log(q_\forall)$, positivity of $(\log(q_\forall))^{1/2}$, and $1 \leq 10\delta \log(2\delta \log(q_\forall))$. Lean also verifies the C1 coefficient replacement $80 d_{\mathrm{mod}}/\ell \leq \delta h^{-1/2}$ used immediately after applying Theorem 1.10. Lean also verifies the C1 upper-window replacement for the Theorem 1.10 error term $20(d^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$. Combining these, Lean proves the first displayed post-Theorem-1.10 bound in the proof of Corollary 2.2(ii). Lean also proves the displayed comparison $\frac{1}{6}\log(q_2) - \frac{1}{6}\log(q) \leq \frac{1}{3}h^{1/2}\log(2\delta h)$ from the intermediate $h^{1/2}\log(\ell)$ bounds. Lean then combines this with the bounded $q_\forall/q_2$ discrepancy to prove the displayed pre-ϵ_E bound for $\frac{1}{6}h$. For Corollary 2.2(ii), Lean proves the final $\frac{1}{6}\log(q)$ bound from the displayed $\log(q)$, $\log(q_2)$, and $\log(q_\forall)$ chain. Lean records the proof's definition of ϵ_E and verifies the stated lower estimate $\epsilon_E \geq 5\delta h^{-1/2}$. Lean also verifies the source conversion $(15\delta)^2 h^{1/2}\log(2\delta h) \leq \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$. Combining this with $\delta h^{-1/2} \leq \epsilon_E/5$, Lean proves the preliminary inequality needed for the move-left step. Lean then verifies the algebraic move-left step that introduces the denominator $(1 - 2\epsilon_E/5)^{-1}$. Lean also proves the final ϵ_E-absorption step used just before C2: the intermediate expression with denominator $1 - 2\epsilon_E/5$ is bounded by $(1 + \epsilon_E)$ times the log-degree sum plus C_K. Lean now composes these steps into the final displayed $h = \log(q_\forall)$ bound. Lean also derives the displayed C2 chain from this h-bound and the tripodal-to-curve log-degree comparison. It also proves the two sum inequalities relating the tripodal different and conductor terms to $\log \mathrm{diff}_X + \log \mathrm{cond}_D$. For the Corollary 2.2/2.3 exceptional-set passage, Lean proves that a lower bound off a finite exceptional set and a lower bound on the exceptional set imply a global lower bound, and specializes this to the Theorem A discrepancy. For Corollary 2.3, Lean proves the stated bounded-discrepancy conclusion from an explicit GenEll transfer hypothesis; GenEll itself is not formalized here. Finally, Lean combines the Corollary 2.2(i) bounded-discrepancy height comparison with the C2 bound to construct the Theorem A lower-bound shadow.

7 Missing Mathematics

The present code does not yet formalize:

- initial Θ-data as in IUT I;
- actual Hodge theaters, Frobenioids, prime strips, log-shells, or Θ-links;
- the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
- the construction of the actual theta values, Gaussian/splitting monoids, $F^{\times}_{\mathrm{mod}}$-actions, Kummer isomorphisms, mono-analyticization, and Frobenioids behind the present shadows;
- the actual construction of log-shells and tensor packets attached to mono-analytic prime-strips in IUT III, Theorem 3.11;
- the genuine holomorphic hull and determinant operations of Step (xi), beyond the present finite determinant/upper-ray/two-computation/tensor-power shadows;
- the global realified Frobenioids of Step (xii), beyond the present integer log-volume shift model;

- the arithmetic-divisor and local log-volume construction behind the IUT IV Theorem 1.10 numerical estimates, and the Diophantine applications;
- the construction of the normalized height, log-different, and log-conductor functions used in IUT IV, Theorem A;
- the construction of the bounded-discrepancy equivalences asserted in IUT IV, Corollary 2.2(i);
- a proof that the current Hodge/SHE transport obligations follow from Mochizuki's full hypotheses.

8 Next Work

The next significant formal step is not another wrapper. It is to replace current explicit Hodge/SHE transport assumptions by constructions closer to the papers:

$$\begin{gathered}\text{theta values } q^{j^2} \\ \rightsquigarrow \text{ Gaussian and splitting monoids} \\ \rightsquigarrow \text{ square/full-label preservation} \\ \rightsquigarrow \text{ hull/log-volume route.}\end{gathered}$$

Only after this bridge is formalized can the first-pass experiment begin to test the actual mathematical load-bearing part of Mochizuki's response to Scholze–Stix.

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